

AGROSPEAR CROPCARE CO., LTD.

IN-PROCESS QUALITY CONTROL (IPQC) INSPECTION REPORT

氯虫苯甲酰胺 200g/L 悬浮剂 (SC) 生产制程过程巡检检验报告

Report No.: AGR-IPQC-INS-20260826-02
Inspection Date: 2026-08-26
Production Line: SC Formulation Workshop Line #2
Execution Standard: SOP-QC-AGR-004 / CIPAC

Product Name: Chlorantraniliprole 200g/L SC	Batch Number: AS-INS-20260826-03	Batch Size: 12,000 Liters	Process Status: CONTROLLED / 制程受控
TC Raw Material Lot: TC-20260815-98 (Purity 98.42%)	IPQC Inspector: Sun Qian (孙倩)	Shift / Line Supervisor: Li Jianguo (李建国)	QA Gate Clearance: IPQC GATE 2 RELEASED

STAGE 2.1: PRE-DISPERSION & SLURRY MIXING / 预混合搅拌与初浆配制

Process Item / 工序	Control Parameters & Specs	Actual Monitored Data	Sampling Method	Verdict
Raw Material Dosing 原药与助剂投料核对	• TC Weight: 2,438.5 kg (Acc. ±0.1%) • Wetting/Dispersing agent & Antifreeze double check	• TC Actual: 2,438.4 kg • Dispersant & Solvents verified	Digital Load Cell Printout & Barcode Scan	Pass
High-Shear Mixing 高速剪切分散工况	• Agitator Speed: 800 – 1200 RPM • Mixing Time: 30 – 45 min • Slurry Temperature < 40.0 °C	• Speed: 1050 RPM • Duration: 38 min • Slurry Temp: 32.8 °C	Continuous Sensor & Infrared Thermometer	Pass
Slurry Homogeneity 初浆均相状态初检	• No agglomerates or un-wetted TC lumps • Pre-milling Fineness D90 < 35 µm	• Uniform creamy suspension • Pre-milling D90: 22.4 µm	Microscopy & Laser Diffraction (100-mesh sieve)	Pass

STAGE 2.2: WET BEAD MILLING & FINENESS MONITORING / 卧式砂磨研磨与细度监控

Milling Stage / 阶段	Critical Process Parameters (CPP)	In-line Measured Value	Frequency / Instrument	Verdict
Primary Milling (Pass 1) 一段粗磨监控	• 1.0mm Zirconia Beads load: 85% • Chamber Outlet Temp ≤ 38.0 °C • Target D90 < 10.0 µm	• Outlet Temp: 33.6 °C • Pressure: 0.12 MPa • Particle Size D90: 7.82 µm	Every 45 min Malvern Mastersizer	Pass
Fine Milling (Pass 2) 二段超细研磨监控	• 0.8mm Yttrium Zirconia Beads load: 85% • Chamber Outlet Temp ≤ 38.0 °C • Chamber Pressure < 0.25 MPa • Target Fineness: D50 < 2.0µm, D90 < 4.5µm	• Outlet Temp: 32.4 °C (Peak: 34.8°C) • Pressure: 0.14 MPa • D50: 1.38 µm • D90: 2.85 µm	Every 30 min Malvern Mastersizer	Pass
Grinding Wear Check 研磨介质破损与磨损	• Bead Wear Rate < 0.1% per batch • Screen gap: 0.3mm without clogging	• Screen differential pressure: 0.02 MPa • No bead fragments detected	Differential Pressure Gauge & Sieve Check	Pass

STAGE 2.3: STRUCTURING, VISCOSITY & WET SIEVING / 调配增稠、黏度与湿筛控制

Test Item / 检测项目	In-Process Specification Limit	Test Result / 实测结果	Standard Method	Verdict
Thickener Pre-gel Addition 黄原胶/硅酸镁铝胶体调配	Pre-hydrated colloid fully dispersed; Low-shear mixing 200–400 RPM for 60 min	Agitation 300 RPM, 60 min complete dispersion	Operational Timer & PLC log	Pass
Apparent Viscosity / 黏度	400 – 1200 mPa·s (at 20°C)	760 mPa·s	Brookfield Viscometer (#2 / 30 RPM)	Pass
Thixotropic Index / 触变比	Ratio (6 RPM / 60 RPM) ≥ 2.5	3.10 (Good anti-settling structure)	Brookfield Viscometer Ratio	Pass
Wet Sieve Test (75µm sieve) 中间体湿筛试验	≥ 99.0 % through (no coarse agglomerates)	99.92 % through	CIPAC MT 185	Pass
Active Content Quick Check 中间体含量快速测定	195.0 – 208.0 g/L (Before final release)	201.2 g/L	HPLC In-Process Check	Pass

STAGE 2.4: AUTOMATED PACKAGING LINE INSPECTION / 自动灌装、封口与喷码巡检

Inspection Item / 项目	Standard Parameter / 标准	Monitored Sampling (Every 30 min)	Sampling Frequency	Verdict
Net Volume / 净含量	1000 ± 5.0 mL (100% in-line dynamic check)	Average: 1002.4 mL (Min: 998.2, Max: 1004.1)	100% Weigher + 5 bottles/ 30m manual volumetric	Pass
Capping Torque / 瓶盖扭矩	2.5 – 3.5 N·m	2.8 – 3.1 N·m	5 bottles every 30 min (Torque Tester)	Pass
Induction Foil Seal Integrity 电磁感应铝箔封口密封性	100% hermetic seal; Vacuum chamber -70 kPa for 1 min without bubble/leak	0/50 leaking bottles under -70 kPa vacuum	10 bottles every 1 hour (Vacuum Leak Tester)	Pass
2D QR Laser Batch Coding 激光赋码与追溯清晰度	Legible batch code, EXP date, and 100% 2D QR readability matching ERP	Batch: AS-INS-20260826-03; QR 100% verified	Continuous Machine Vision + 10 btl/s/1h check	Pass

SECTION 5: IPQC RELEASE DISPOSITION & SIGN-OFF / 制程放行决议与签批

IPQC PROCESS RELEASE DISPOSITION / 制程放行综合结论：

All process parameters across Pre-mixing, Bead Milling (D90 = 2.85 μ m, Temp = 32.4°C), Colloid Structuring (Viscosity = 760 mPa·s), and Automated Packaging (Foil seal vacuum 100% passed) strictly comply with Enterprise Standard Q/AS-0826 and SOP-QC-AGR-004. **The batch is successfully released from Stage 2 (IPQC) and transferred to Stage 3 (FQC) for official COA testing and quarantine archiving.**

本批次氯虫苯甲酰胺 200g/L 悬浮剂在预混、砂磨细化 (D90 = 2.85 μ m, 研磨温度 32.4°C)、调配增稠 (黏度 760 mPa·s) 及自动灌装封口 (-70 kPa 铝箔密封全合格) 等关键制程控制点均符合标准规程, 准予第二阶段 (IPQC) 制程放行, 进入第三阶段 (FQC) 全项终检与成品放行阶段。

IPQC Inspector / 制程巡检员	Workshop Lead / 车间主任	Process Engineer / 工艺工程师	QA IPQC Manager / 质量主管审批
Sun Qian (孙倩) Date: 2026-08-26 Sign: [Sun Qian]	Li Jianguo (李建国) Date: 2026-08-26 Sign: [Li Jianguo]	Zhang Wei (张伟) Date: 2026-08-26 Sign: [Zhang Wei]	Dr. Alex Chen (陈博士) Date: 2026-08-26 Sign: [Alex Chen] [PASSED]